Synthetic Instruments by Adversarial Splitting: Identification through Non-Gaussianity*

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Abstract

We study the estimation of a linear structural coefficient when a regressor is endogenous and no credible external instrument is available. The idea is to split the endogenous regressor into an exogenous component and a remainder, $x = x_{\text{exo}} + r$ with $x_{\text{exo}} \perp\!\!\!\perp u$, and to use x_{exo} as an internally generated ("synthetic") instrument or, equivalently, r as a control function. We first explain why a naive version of this idea—generating many candidate instruments with a generative network trained to pass weak-instrument and over-identification tests—cannot work: exogeneity never enters a computable criterion. We then make the identifying assumption explicit: the regressor and the structural error are driven by mutually independent sources, at least two of which are non-Gaussian. Under this assumption the structural coefficient is identified from fourth-order cumulants up to a two-point set $\{\beta, \beta + \lambda\}$, a label ambiguity that is intrinsic to the model and that we resolve with a sign restriction or an auxiliary variable. The estimation layer is a neural network that learns the split under a Hilbert–Schmidt independence penalty, trained on one half of the sample, with estimation and weak-identification-robust inference on the other half. We also show that, in the over-complete linear model, the exogenous source is generally *not* a measurable function of the data, so the network can only target moment validity rather than exact independence; a pilot simulation confirms that the unanchored split collapses towards OLS while the proposed diagnostics correctly report the failure. We propose a complete battery of pre-estimation, relevance, exogeneity and sensitivity tests, and a Monte Carlo design that pits the method against OLS, an oracle instrument, structural error projection, heteroskedasticity-based instruments, Gaussian-copula control functions and a linear cumulant (over-complete ICA) estimator.

Keywords: endogeneity, internal instruments, independent component analysis, higher-order cumulants, HSIC, control function, weak identification.

1 Introduction

Consider the linear structural model

$$y = x\beta + w'\delta + u, \qquad \mathbb{E}[xu] \neq 0, \tag{1}$$

where x is a scalar endogenous regressor, w is a vector of exogenous covariates and u is the structural error. The textbook solution is an external instrument: a variable that moves x and is excluded from (1). When no such variable is credible, researchers turn to *internal* instruments, built from the data

*Research proposal and Monte Carlo testbed. Code, tests and simulation scripts accompany this paper in the `ivsplit` package (`src/ivsplit`, `tests`, `scripts`). The literature search is not exhaustive.

at hand by exploiting heteroskedasticity (Rigobon, 2003; Lewbel, 2012), higher moments (Lewbel, 1997; Erickson and Whited, 2002), latent discrete structure (Ebbes et al., 2005), distributional assumptions (Park and Gupta, 2012) or functional form (Dong et al., 2026).

This paper develops a variant of that programme. We seek a decomposition

$$x = x_{\text{exo}} + r, \qquad x_{\text{exo}} \perp\!\!\!\perp u, \tag{2}$$

and use x_{exo} as the source of exogenous variation. Two ingredients define the proposal:

(i) **Identification** comes from mutually independent, non-Gaussian sources of variation, as in independent component analysis (ICA; Comon, 1994) and its causal counterparts with latent variables (Hoyer et al., 2008; Salehkaleybar et al., 2020; Wang and Drton, 2023; Tramontano et al., 2025).

(ii) **Estimation** uses a neural network that learns the split, with independence enforced by a Hilbert–Schmidt independence criterion (HSIC; Gretton et al., 2005, 2008) in the spirit of HSIC-X (Saengkyongam et al., 2022), and with sample splitting so that inference does not reuse the data that shaped the instrument.

Our contributions are fourfold. First, we state precisely what has to be assumed: any synthetic instrument replaces the exclusion restriction of an external instrument with another untestable hypothesis, here independence of latent sources (Section 2). Second, we give a self-contained identification result for the canonical latent-confounder model, which shows that the coefficient is identified from fourth-order cumulants only up to the set $\{\beta, \beta + \lambda\}$ and that this ambiguity is a genuine observational equivalence, not an artefact of the estimator (Section 3). Third, we show that the exogenous source is generally not a measurable function of observables in this over-complete setting, which sharpens what a network can and cannot learn and motivates an anchored version of the estimator (Section 4). Fourth, we provide a complete testing battery and a Monte Carlo design, with an open-source implementation and unit tests (Sections 5–7).

2 Why the naive synthetic-instrument idea fails

The first version of this project generated $2k$ candidate instruments with a generative adversarial network trained to pass weak-instrument and Sargan tests, and filtered the excess with Hausman tests. The design fails for reasons that are worth recording because they constrain every variant of the idea.

1. *Exogeneity is not in the objective.* The condition $\text{Cov}(z, u) = 0$ involves the unobserved u and appears in no computable criterion. A search over candidates converges to *strong* instruments, not to *valid* ones.

2. *Passing an over-identification test is necessary, not sufficient.* Instruments manufactured from the same data inherit the same contamination and agree with each other on the same wrong value; the Sargan–Hansen test (Sargan, 1958; Hansen, 1982) has no power against a common bias.

3. *Training to pass a test is selection on the test outcome.* It produces pre-test bias, a winner's curse and a severe multiple-testing problem across $2k$ candidates.

4. *The Hausman test answers a different question.* Hausman (1978) compares OLS with IV and tests the endogeneity of the regressor. Testing a subset of instruments calls for the difference-in-Hansen (C) statistic (Eichenbaum et al., 1988).

5. *No algorithm creates identification.* If the data contain no source of exogenous variation, no transformation of them will. Every synthetic instrument trades the exclusion restriction of an external instrument for another untestable hypothesis.

The question therefore becomes: *which identifying hypothesis is the researcher willing to maintain in order to perform the split* (2)*?* We answer it with independence and non-Gaussianity of latent sources.

3 Model and identification

3.1 The canonical latent-confounder model

Let $\tilde{x} = x - \mathbb{E}[x \mid w]$ and $\tilde{y} = y - \mathbb{E}[y \mid w]$ denote the regressor and outcome after partialling out the exogenous covariates. Our leading structure is

$$x = g(w) + H + e_X, \qquad u = \lambda H + e_Y, \qquad y = \beta x + w'\delta + u, \tag{3}$$

so that $\tilde{x} = H + e_X$ and $\tilde{y} = \beta\tilde{x} + u = (\beta + \lambda)H + \beta e_X + e_Y$. Here H is a latent confounder, e_X is the exogenous variation in x that we wish to recover, and e_Y is idiosyncratic outcome noise. All three are mean zero and independent of w.

3.2 Assumptions

Assumption H1 (Independent sources)**.** *H, e_X and e_Y are mutually independent and independent of w.*

Assumption H2 (Non-Gaussianity)**.** *The fourth cumulants $\kappa_H = \mathrm{cum}_4(H)$ and $\kappa_X = \mathrm{cum}_4(e_X)$ are non-zero.*

Assumption H3 (Class restriction)**.** *The map from sources to observables belongs to a class for which the independent decomposition is unique up to the symmetries made explicit below: linear mixing after partialling out w as in* (3)*, or a non-linear class made identifiable by additive noise, sparsity of the mixing map or an observed auxiliary variable (Hyvärinen et al., 2019; Khemakhem et al., 2020; Zheng et al., 2022).*

Assumption H4 (Relevance)**.** *The exogenous component has non-zero variance and is correlated with x:* $\mathrm{Var}(e_X) > 0$.

Assumption H5 (Labelling)**.** *One of the following holds.* ***(L1)*** *The sign of the confounding,* $\mathrm{sign}(\lambda)$*, is known.* ***(L2)*** *There is an auxiliary variable a (a component of w or an additional observable) such that the law of H does not depend on a while* $\mathrm{Var}(e_X \mid a)$ *does.*

Assumption H1 is a disguised exclusion restriction and cannot be tested. Assumption H2 is checkable. Assumption H3 is necessary because non-linear ICA is not identifiable in general (Hyvärinen and Pajunen, 1999). Assumption H5 is needed because of the symmetry established next.

3.3 Identification up to a two-point set

Write $c_{4-k,k} = \operatorname{cum}(\underbrace{\tilde{x},\dots,\tilde{x}}_{4-k},\underbrace{\tilde{y},\dots,\tilde{y}}_{k})$ for the joint fourth-order cumulants, and abbreviate $c_k \equiv c_{4-k,k}$. By multilinearity of cumulants and independence of the sources,

$$c_k = \kappa_H(\beta+\lambda)^k + \kappa_X\beta^k + \kappa_Y\,\mathbf{1}\{k=4\}, \qquad k=0,\dots,4. \tag{4}$$

For $k \le 3$ this is a two-term exponential sum in the unknown slopes $s_H = \beta+\lambda$ and $s_X = \beta$, so it satisfies the linear recursion of Prony's method:

$$c_{k+2} = p\,c_{k+1} + q\,c_k, \quad k=0,1, \qquad p = s_H + s_X,\ q = -s_H s_X. \tag{5}$$

Proposition 1 (Identification up to labels)**.** *Under Assumptions H1–H2 in model (3) with $\lambda \neq 0$:*

(a) The matrix $M = \begin{pmatrix} c_1 & c_0 \\ c_2 & c_1 \end{pmatrix}$ has $\det M = -\kappa_H\kappa_X\lambda^2 \neq 0$, so (p,q) is identified from $M(p,q)' = (c_2,c_3)'$, and the set $\{\beta,\beta+\lambda\}$ is identified as the roots of $t^2 - pt - q = 0$.

(b) The set is sharp: the model with parameters $(\beta',\lambda') = (\beta+\lambda,-\lambda)$ and sources $(H',e_X') = (e_X,H)$ generates exactly the same distribution of $(\tilde{x},\tilde{y})$.

Proof. (a) Substituting (4), $\det M = (\kappa_H s_H + \kappa_X s_X)^2 - (\kappa_H s_H^2 + \kappa_X s_X^2)(\kappa_H+\kappa_X) = -\kappa_H\kappa_X(s_H - s_X)^2 = -\kappa_H\kappa_X\lambda^2$. The recursion (5) follows from $s_j^{k+2} = ps_j^{k+1} + qs_j^k$ for $j \in \{H,X\}$, which holds because each s_j solves $t^2 - pt - q = 0$. (b) Under the alternative parameters, $\tilde{x} = H' + e_X' = e_X + H$ and $\tilde{y} = (\beta'+\lambda')H' + \beta' e_X' + e_Y = \beta e_X + (\beta+\lambda)H + e_Y$, which coincide with the original variables. □

Remark 1. Part (b) is the population version of the "identified only up to two points" result of Delbianco (2026) and of the label ambiguity in latent-variable LiNGAM (Hoyer et al., 2008; Tramontano et al., 2025). It is not a weakness of any particular estimator: without Assumption H5 *no* procedure can tell the confounder from the exogenous source, because both enter x with unit loading.

Proposition 2 (Labelling)**.** *Under the conditions of Proposition 1: (L1) if $\lambda > 0$ then β is the smaller root, and if $\lambda < 0$ the larger; (L2) β equals the slope of $\operatorname{Cov}(\tilde{x},\tilde{y} \mid a)$ on $\operatorname{Var}(\tilde{x} \mid a)$ across values of a.*

Proof. (L1) is immediate. For (L2), $\operatorname{Var}(\tilde{x} \mid a) = \sigma_H^2 + \sigma_X^2(a)$ and $\operatorname{Cov}(\tilde{x},\tilde{y} \mid a) = (\beta+\lambda)\sigma_H^2 + \beta\sigma_X^2(a)$, so the two conditional moments are affine in $\sigma_X^2(a)$ with slope ratio β. □

Remark 2. Under (L2) the second-moment argument alone identifies β, without any appeal to non-Gaussianity; it is identification through heteroskedasticity (Rigobon, 2003; Lewbel, 2012). In our framework (L2) is used only to *pick* one of the two roots, which needs far less than consistent estimation of the slope; the cumulant equations then provide an over-identifying check.

Proposition 3 (Gaussian non-identification)**.** *If H, e_X and e_Y are Gaussian, β is not identified: the distribution of $(\tilde{x},\tilde{y})$ depends on the five parameters $(\beta,\lambda,\sigma_H^2,\sigma_X^2,\sigma_Y^2)$ only through the three moments $\operatorname{Var}(\tilde{x})$, $\operatorname{Cov}(\tilde{x},\tilde{y})$ and $\operatorname{Var}(\tilde{y})$, and every β' in a non-degenerate interval containing β is observationally equivalent.*

Proof. Gaussian vectors are characterised by their covariance. Fix β' near β and keep $\sigma_H'^2 = \sigma_H^2$, $\sigma_X'^2 = \sigma_X^2$. Choose λ' so that $\operatorname{Cov}(\tilde{x},\tilde{y}) = \beta'\operatorname{Var}(\tilde{x}) + \lambda'\sigma_H^2$ and $\sigma_Y'^2$ so that $\operatorname{Var}(\tilde{y})$ is matched; by continuity $\sigma_Y'^2 > 0$ for β' close enough to β. □

Proposition 3 is the population reason why scenario 3 of the Monte Carlo is a negative control: an honest procedure must report weak or failed identification there, rather than a confident estimate. It also explains why the identification-strength statistic of Section 3.4 is built on $\det M$.

3.4 An identification-strength diagnostic

Proposition 1 shows that identification hinges on $\det M = -\kappa_H \kappa_X \lambda^2$, which vanishes when either source is mesokurtic (e.g. Gaussian) or when there is no confounding. We report

$$D = \frac{\widehat{\det M}^{\,2}}{\widehat{\mathrm{Var}}_{\mathrm{boot}}\big(\widehat{\det M}\big)}, \tag{6}$$

a Wald statistic for $H_0 : \det M = 0$, with a nonparametric bootstrap variance. Small values of D (below the χ^2_1 critical value) indicate that the higher-order information is insufficient to separate the sources, in the spirit of the D diagnostic of Delbianco (2026). When $\lambda = 0$ a small D is benign—OLS is then consistent—and should be read together with a Hausman test.

3.5 The linear identification layer as an estimator

Replacing population cumulants by sample cumulants of the w-residualised data, solving the (possibly over-identified, if third-order cumulants are added) linear system for $(\hat{p}, \hat{q})$ and labelling the roots with (L1) or (L2) gives a simple closed-form estimator, which we call *Cumulant-ICA*. It is the over-complete (2×3) ICA estimator of the canonical model and is closely related to the higher-moment estimators of Lewbel (1997) and Erickson and Whited (2002). We use it both as a competitor and as an anchor for the network (Section 4.4).

4 The neural split

4.1 Procedure

Step 1. Sample splitting. Randomly split the sample into halves A and B. All learning happens on A; $\hat{\beta}$, standard errors and every test are computed on B.

Step 2. Partialling out. Compute $\tilde{x}, \tilde{y}$ by regressing x, y on w (linearly, with a sieve, or with cross-fitted machine learning as in Chernozhukov et al., 2018).

Step 3. Training. A network f_θ maps $(\tilde{x}, w)$ to x_{exo} and sets $r = \tilde{x} - x_{\mathrm{exo}}$. The parameters minimise

$$\mathcal{L}(\theta, \beta) = \alpha\ \mathrm{HSIC}(x_{\mathrm{exo}}, r) + \gamma\ \mathrm{HSIC}(\tilde{y} - \beta\tilde{x},\ x_{\mathrm{exo}}) + \eta\left[(c_{\min} - \rho^2)_+ + (\rho^2 - c_{\max})_+\right], \tag{7}$$

where $\rho = \mathrm{Corr}(x_{\mathrm{exo}}, \tilde{x})$. The first term imposes independence between the two parts of the split; the second is the HSIC-X criterion of Saengkyongam et al. (2022) with the learned instrument; the third is a relevance band that rules out degenerate solutions. The reconstruction term $\|x - x_{\mathrm{exo}} - r\|^2$ of the original proposal vanishes by construction. β is updated in alternation by the IV formula $\beta \leftarrow \sum x_{\mathrm{exo}}\tilde{y} / \sum x_{\mathrm{exo}}\tilde{x}$.

Step 4. Estimation. On B, compute x_{exo} and r and estimate β by 2SLS with x_{exo} as the excluded instrument ("split-IV"), or by OLS of y on (x, w, r) ("split-CF").

Step 5. Inference. Report heteroskedasticity-robust standard errors, the robust first-stage F (Montiel Olea and Pflueger, 2013), and the Anderson–Rubin confidence set (Anderson and Rubin, 1949), which remains valid when the learned instrument is weak.

Step 6. Diagnostics. Report D from (6) on B next to $\hat{\beta}$, and the battery of Section 5.

4.2 Degenerate solutions

Three degenerate solutions must be excluded. (i) $x_{\text{exo}} \approx$ constant is independent of everything but irrelevant; the relevance floor $c_{\min}$ excludes it. (ii) $x_{\text{exo}} = \tilde{x}$ has maximal relevance and, with $r \equiv 0$, trivially satisfies $\text{HSIC}(x_{\text{exo}}, r) = 0$, but the IV estimator collapses to OLS; the relevance cap $c_{\max} < 1$ excludes it. (iii) Over-fitting on the training sample inflates the first-stage F and recreates the many-instruments bias; sample splitting neutralises it.

4.3 What the network can and cannot recover

A natural reading of (2) is that the network should output $x_{\text{exo}} = e_X$. The next result shows that this is generally impossible in the over-complete model, where three sources drive two observables.

Proposition 4 (No measurable exact split)**.** *In model (3) under Assumption H1, suppose $\lambda \neq 0$, the characteristic function of e_Y has no real zeros, H has support $\mathbb{R}$, and the characteristic function of e_X has no real zeros. Then every measurable f with $f(\tilde{x}) \perp\!\!\!\perp u$ is almost surely constant.*

Proof sketch. Since e_Y is independent of (H, e_X) and its characteristic function does not vanish, $f(\tilde{x}) \perp\!\!\!\perp \lambda H + e_Y$ implies $f(\tilde{x}) \perp\!\!\!\perp H$ by deconvolution of the joint characteristic function. For any Borel set A, independence then requires $\Pr(H + e_X \in A \mid H = h) = \int \mathbf{1}_A(h + t)\, dF_{e_X}(t)$ to be constant in h on the support of H, i.e. the convolution of $\mathbf{1}_A$ with the law of $-e_X$ is constant on $\mathbb{R}$. Taking Fourier transforms and using that the characteristic function of e_X has no zeros, by Wiener's Tauberian theorem $\mathbf{1}_A$ is a.e. constant, so the σ-field generated by $f(\tilde{x})$ is trivial. □

Laplace, Gaussian and centred exponential laws all satisfy the conditions, and adding w to the inputs does not help when w is independent of the sources. Two consequences follow.

- The population objective (7) cannot be driven to zero, and its minimiser in (θ, β) need not be at the true β. In particular, at the OLS slope the residual $\tilde{y} - \beta_{\text{OLS}}\tilde{x}$ is uncorrelated with $\tilde{x}$, so the HSIC-X term only penalises higher-order dependence, which is weak in moderate samples. Our pilot simulations (Section 7.4) confirm that the unanchored split converges near OLS.
- What *is* attainable is *moment* validity. Writing $m(\tilde{x}) = \mathbb{E}[H \mid \tilde{x}]$, any function f with $\mathbb{E}[f(\tilde{x})m(\tilde{x})] = 0$ and $\mathbb{E}[f(\tilde{x})\tilde{x}] \neq 0$ is a valid and relevant instrument, since $\mathbb{E}[f(\tilde{x})u] = \lambda\mathbb{E}[f(\tilde{x})m(\tilde{x})]$. Such functions exist in abundance (they form a subspace of codimension one), but selecting one requires knowledge of m, which in turn depends on the laws of H and e_X—exactly what the cumulant layer identifies up to labels.

There are also non-generic cases in which an exact split exists, for example when H is discrete and e_X has bounded support with non-overlapping translates; such structures are the province of latent-mixture approaches (Ebbes et al., 2005).

4.4 Anchored split

Proposition 4 motivates a revised estimation layer that lets the identification layer do the identifying and the network do the approximating:

(i) On sample A, estimate β (and, if desired, the source laws) with the cumulant layer and label the roots with (L1) or (L2), obtaining $\hat{\beta}_A$.

(ii) Train f_θ on A with β fixed at $\hat{\beta}_A$ in the HSIC-X term, so that the network searches for instrument functions $f(\tilde{x}, w)$ whose implied moment conditions are approximately satisfied at the identified value, and use the relevance band to keep them informative.

(iii) On B, estimate β by 2SLS with the learned functions and test the over-identifying restrictions they imply.

The non-linear value of the network lies where the linear layer is silent: a non-linear first stage $g(w)$, auxiliary variables that modulate the law of e_X (Assumption H3), and efficient combination of moment conditions. Whether the anchored split improves on the plain cumulant estimator in finite samples is an empirical question that the Monte Carlo is designed to answer.

5 Testing battery

All tests are computed on the estimation half B. Table 1 summarises them.

Exogeneity cannot be tested directly. Because u is unobserved, every exogeneity test uses the estimated residual $\hat{\varepsilon} = y - x\hat{\beta} - w'\hat{\delta}$ with $\hat{\beta}$ computed from x_{exo} itself. The linear correlation between x_{exo} and $\hat{\varepsilon}$ is therefore zero by construction and only non-linear dependence is informative. Since $\hat{\beta}$ is estimated, permutation or naive bootstrap calibration is only approximate; a bootstrap that re-estimates β in each draw is preferable.

Over-identification with functions of the instrument. If x_{exo} is contaminated, the moment conditions $\mathbb{E}[f_j(x_{\text{exo}})(y - x\beta - w'\delta)] = 0$ generally imply different values of β for different f_j, and Hansen's J rejects. Two blind spots deserve emphasis, both confirmed by the accompanying unit tests: under Gaussianity $\mathbb{E}[u \mid x_{\text{exo}}]$ is linear and all f_j imply the same wrong β; and if H and e_X are *exchangeable* (identically distributed), then $\mathbb{E}[H \mid H + e_X] = (H + e_X)/2$ is again linear and the test has no power against the fully contaminated instrument $\tilde{x}$. The test has power when the conditional expectation of the confounder given the instrument is non-linear.

Procedural safeguards.

- *Circularity.* If the network minimised HSIC$(\hat{\varepsilon}, x_{\text{exo}})$, testing the same quantity on the same data tests what was optimised. Train on A, test on B.
- *Gaussianity.* Under normality, independence reduces to zero correlation, which holds by construction, and every independence test passes vacuously. Always report the non-Gaussianity diagnostics alongside.
- *Multiple testing.* When several exogeneity tests are reported, adjust with Bonferroni or Holm (1979).
- *Post-selection inference.* Hyper-parameters and architecture choices informed by test outcomes must be made on A.

Table 1: Tests applied before and after estimation.

Test	What it checks
Pre-estimation (identification conditions)	
Skewness, kurtosis, Jarque–Bera (Jarque and Bera, 1987)	Assumption H2, on $\tilde{x}$ and on OLS residuals
Identification strength D, eq. (6)	Whether the higher-order information separates the sources
Breusch–Pagan / White (Breusch and Pagan, 1979; White, 1980)	Heteroskedasticity, needed for the Lewbel competitor and for (L2)
Relevance	
Robust first-stage F / Kleibergen–Paap (Montiel Olea and Pflueger, 2013; Kleibergen and Paap, 2006)	Strength of x_{exo}; computed on B only
Anderson–Rubin set (Anderson and Rubin, 1949)	Weak-instrument-robust inference on β
Exogeneity of x_{exo} (the critical point)	
HSIC / distance covariance of x_{exo} and $\hat{\varepsilon}$ (Gretton et al., 2008; Székely et al., 2007)	Dependence beyond correlation (which is zero by construction)
Hansen J with $f_1(x_{\text{exo}}), f_2(x_{\text{exo}}), \ldots$ (Hansen, 1982)	Contaminated instruments imply different βs across functions
C test (Eichenbaum et al., 1988)	Validity of a subset of instruments
Agreement between strategies	Hausman-type contrast with Lewbel, SP, copula, Cumulant-ICA
Hausman, OLS vs IV (Hausman, 1978)	Is there endogeneity to correct?
Negative-control outcomes	Zero effect on an outcome x cannot affect
HSIC(x_{exo}, r)	Independence of the split (necessary only)
Sensitivity	
Conley et al. (2012); Nevo and Rosen (2012)	Range of $\hat{\beta}$ if $\text{Cov}(x_{\text{exo}}, u) \neq 0$

6 Related literature

Table 2 positions the proposal. Two strands are closest. Dong et al. (2026) also split x without an external instrument, constructing $z = x - P_S x$ by projecting x on a sieve approximation of the confounding $\mathbb{E}[\varepsilon \mid x]$; identification there comes from functional form (the confounding has no linear part in x), which fails under joint normality and does not cover linear confounding. Delbianco (2026) tests the exclusion restriction of a single instrument with fourth-order cumulants under a latent non-Gaussian factor, an over-complete 2×3 ICA problem; the direct effect is identified only up to two points, which is the counterpart of our Proposition 1(b), and power requires substantial kurtosis and n in the low thousands.

Apparent gap. A non-linear split of the endogenous regressor, learned by adversarial or HSIC independence, identified by non-Gaussianity, and used as an instrument or control function with valid inference. The linear identification layer exists already; the defensible contribution is the non-linear estimation layer with inference and Monte Carlo evidence. Referees will likely describe the method as "over-complete ICA by another name"; Proposition 4 and the anchored split are our answer to what the network adds.

Table 2: Position in the literature.

Work	What it does	Relation to this proposal
Dong et al. (2026)	Internal instrument $z = x - P_S x$; identification by functional form; LASSO over a basis; non-parametric Hausman-type test	Splits x without IV; different identification; no networks; fails under joint normality and linear confounding
Delbianco (2026)	Tests exclusion with one instrument via fourth-order cumulants; over-complete ICA; effect identified up to two points	Shows e_X identifiable under latent confounding; needs Z, linear, power only with kurtosis and large n; provides a D diagnostic
Tramontano et al. (2025)	Causal effects in lvLiNGAM from higher-order cumulants with one proxy or instrument	Close neighbour of the identification layer
Hoyer et al. (2008); Salehkaleybar et al. (2020); Wang and Drton (2023); Silva and Shimizu (2017)	LiNGAM with latent variables; search for valid instruments under non-Gaussianity	Main risk to the novelty of the identification layer
Saengkyongam et al. (2022)	HSIC-X: minimise HSIC between residuals and instruments	Source of the independence loss; still requires instruments
Hyvärinen et al. (2019); Khemakhem et al. (2020); Zheng et al. (2022)	Identifiable non-linear ICA with auxiliary variables or sparsity	Conditions to make a non-linear split identifiable
Lewbel (1997); Erickson and Whited (2002)	Higher-moment instruments for mismeasured regressors	Ancestors of the cumulant layer
Lewbel (2012); Park and Gupta (2012); Ebbes et al. (2005); Rigobon (2003)	Internal instruments via heteroskedasticity, copulas, latent mixtures, regimes	Competitors and robustness checks
Conley et al. (2012); Nevo and Rosen (2012)	Inference with imperfect exogeneity	Sensitivity analysis
Hartford et al. (2017); Bennett et al. (2019)	Deep IV and deep GMM	Neural IV estimators that assume an external instrument

7 Monte Carlo design

7.1 Base DGP

The base design is (3) with $w \sim N(0, I_2)$, $\delta = (0.5, 0.5)'$, $\beta = 1$, unit variances for all sources, and g either linear or non-linear (softplus and sine terms). Sources are drawn from Laplace (excess kurtosis 3), centred exponential (skewness 2), a symmetric bimodal mixture (negative excess kurtosis), uniform, or Gaussian (negative control).

7.2 Scenarios

S1. *Non-Gaussian latent factor* (Delbianco's home turf): identification should work. Variants with linear and non-linear g.

S2. *Non-linear confounding without a linear part* (SP's home turf): $x = g(w) + v$, $u = \lambda(v^2 -$

$\mathbb{E}v^2) + e_Y$ with skewed v, so OLS is biased while the confounding is a quadratic function of x. SP should win; the proposal should at least not lose badly, and its diagnostics should flag the misspecification.

S3. *Gaussian sources* (negative control): no non-Gaussian method identifies β; the procedure must report weak identification rather than a confident $\hat{\beta}$.

S4. *Violated independence*: correlation ρ between H and e_X, or a direct channel $u \mathrel{+}= \delta_{\text{excl}} e_X$. Measures bias as the hypothesis breaks.

S5. *Placebo and size*: $\lambda = 0$ (exogenous x) and pure-noise candidate instruments. The procedure must not invent exogenous variation nor "pass" tests with garbage.

Aux. *Auxiliary heteroskedasticity*: $\mathrm{Var}(e_X \mid w_1)$ varies with w_1, which enables labelling (L2) and gives the Lewbel competitor its best case.

7.3 Grid, estimators and metrics

The full grid uses $n \in \{500, 1000, 2000, 5000, 10000\}$, $\lambda \in \{0.2, 0.5, 1\}$ and 500–1000 replications per cell. Estimators: OLS; 2SLS with the planted exogenous variation e_X as an oracle instrument (performance ceiling); SP (sieve + LASSO); Lewbel; Gaussian-copula control function; Cumulant-ICA with labelling (L1) and (L2); and the neural split (IV and control-function versions). Delbianco's test is run only in an auxiliary design with a planted instrument. Metrics: bias, median bias and RMSE of $\hat{\beta}$; coverage of nominal 95% intervals (Wald, bootstrap or Anderson–Rubin); size and power of each test of Section 5 against planted violations; and the share of replications in which the method declares that β is not identified.

7.4 Pilot results

Tables 3–5 report a small pilot run (scripts `run_montecarlo.py` and `make_tables.py`). Table 3 covers all scenarios with the closed-form estimators at $\lambda = 0.5$ (S2 uses $\lambda = 0.3$ and S5 uses $\lambda = 0$). Table 4 contrasts S1 and S3 at $\lambda = 1$ for n up to 10000. Table 5 runs the neural split at $n = 2000$. Labelling (L1) assumes $\lambda > 0$ is known, and (L2) uses w_1 as the auxiliary variable. The full grid is left for the next stage. In all tables, Bias is the median bias and Cov. is the coverage of the nominal 95% interval (Wald with robust standard errors; percentile bootstrap for Cumulant-ICA). NI is the share of replications in which the estimator reports that β is not identified: D below the χ^2_1 95% critical value, and for the split also $F < 10$ or an unbounded Anderson–Rubin set.

Reading the pilot. Six patterns stand out.

(i) The identification layer separates S1 from S3. With $\lambda = 1$ (Table 4), Cumulant-ICA with labelling (L1) is essentially median-unbiased in S1. Its RMSE falls from 0.30 at $n = 1000$ to 0.07 at $n = 10000$, coverage is close to nominal, and the share of replications flagged as not identified falls from 64% to 0%. In the Gaussian S3 the same estimator is erratic (RMSE in the units or larger), but D flags 100% of the replications. This is the first test of the research plan, and the layer passes it.

(ii) Identification through higher moments is weak. With $\lambda = 0.5$ (Table 3), $\det M \propto \lambda^2$ is four times smaller. At $n = 1000$ almost every S1 replication is flagged, and at $n = 5000$ a third still are. This matches the sample sizes that Delbianco (2026) needs for power. Without the D filter, the RMSE of Cumulant-ICA is dominated by a few explosive replications in which $\widehat{\det M} \approx 0$.

Table 3: Pilot Monte Carlo, closed-form estimators. 200 replications per cell; $\beta = 1$; λ as in the text unless fixed by the scenario.

		$n = 1000$				$n = 5000$			
Scenario	Estimator	Bias	RMSE	Cov.	NI	Bias	RMSE	Cov.	NI
S1 non-Gaussian factor	OLS	0.252	0.254	0.00	0.00	0.250	0.250	0.00	0.00
	2SLS oracle	0.001	0.034	0.96	0.00	-0.001	0.014	0.97	0.00
	SP	0.252	0.253	0.00	0.00	0.251	0.251	0.00	0.00
	Lewbel	0.266	1.083	0.94	0.00	0.297	1.283	0.94	0.00
	Copula CF	0.256	0.442	0.79	0.00	0.263	0.310	0.68	0.00
	Cumulant-ICA (L1)	-0.131	3.070	0.98	0.98	-0.036	0.484	0.96	0.34
	Cumulant-ICA (L2)	0.238	0.342	0.73	0.97	0.274	0.357	0.76	0.36
S1 non-linear g	OLS	0.206	0.207	0.00	0.00	0.203	0.205	0.00	0.00
	2SLS oracle	-0.000	0.035	0.96	0.00	-0.002	0.014	0.98	0.00
	SP	0.200	0.202	0.00	0.00	0.199	0.197	0.00	0.00
	Lewbel	0.212	1.004	0.97	0.00	0.231	1.349	0.98	0.00
	Copula CF	0.321	0.428	0.75	0.00	0.382	0.425	0.32	0.00
	Cumulant-ICA (L1)	-0.142	11.550	0.99	0.99	0.010	0.433	0.96	0.56
	Cumulant-ICA (L2)	0.257	0.338	0.80	0.99	0.226	0.322	0.85	0.55
S2 non-linear confounding	OLS	0.576	0.601	0.00	0.00	0.596	0.602	0.00	0.00
	2SLS oracle	0.576	0.601	0.00	0.00	0.596	0.602	0.00	0.00
	SP	0.015	0.059	0.98	0.00	0.004	0.021	1.00	0.00
	Lewbel	1.330	2.758	0.45	0.00	1.426	1.742	0.47	0.00
	Copula CF	2.081	2.236	0.00	0.00	2.272	2.339	0.00	0.00
	Cumulant-ICA (L1)	0.290	9.508	0.96	0.88	1.627	6.471	0.84	0.91
	Cumulant-ICA (L2)	1.514	1.587	0.10	0.89	1.630	1.695	0.03	0.92
S3 Gaussian	OLS	0.249	0.251	0.00	0.00	0.250	0.251	0.00	0.00
	2SLS oracle	0.002	0.035	0.95	0.00	0.002	0.016	0.94	0.00
	SP	0.248	0.253	0.00	0.00	0.249	0.250	0.00	0.00
	Lewbel	0.279	1.065	0.97	0.00	0.265	1.828	0.99	0.00
	Copula CF	0.338	0.596	0.88	0.00	0.321	0.531	0.89	0.00
	Cumulant-ICA (L1)	-0.104	3.465	0.99	1.00	0.065	686.988	1.00	1.00
	Cumulant-ICA (L2)	0.228	0.638	0.99	1.00	0.267	0.535	0.99	1.00
S4 $\rho = 0.3$	OLS	0.248	0.251	0.00	0.00	0.251	0.251	0.00	0.00
	2SLS oracle	0.114	0.118	0.01	0.00	0.116	0.117	0.00	0.00
	SP	0.241	0.240	0.00	0.00	0.241	0.241	0.00	0.00
	Lewbel	0.319	0.691	0.90	0.00	0.273	0.742	0.93	0.00
	Copula CF	0.373	0.446	0.58	0.00	0.389	0.412	0.13	0.00
	Cumulant-ICA (L1)	0.043	25.362	0.99	0.98	-0.017	1.809	0.96	0.81
	Cumulant-ICA (L2)	0.316	0.343	0.46	0.99	0.317	0.345	0.45	0.78
S4 direct effect 0.3	OLS	0.403	0.402	0.00	0.00	0.399	0.399	0.00	0.00
	2SLS oracle	0.302	0.303	0.00	0.00	0.300	0.300	0.00	0.00
	SP	0.401	0.400	0.00	0.00	0.399	0.399	0.00	0.00
	Lewbel	0.425	1.090	0.80	0.00	0.405	0.716	0.84	0.00
	Copula CF	0.402	0.498	0.58	0.00	0.395	0.424	0.23	0.00
	Cumulant-ICA (L1)	0.284	17.079	0.99	0.98	0.275	26.412	0.99	0.97
	Cumulant-ICA (L2)	0.419	0.451	0.21	0.99	0.403	0.449	0.15	0.98
S5 placebo ($\lambda = 0$)	OLS	-0.002	0.023	0.94	0.00	-0.001	0.010	0.95	0.00
	2SLS oracle	-0.001	0.033	0.94	0.00	0.002	0.014	0.94	0.00
	SP	-0.002	0.029	0.94	0.00	0.000	0.011	0.94	0.00
	Lewbel	0.015	0.572	1.00	0.00	-0.019	0.688	1.00	0.00
	Copula CF	0.010	0.297	0.91	0.00	-0.011	0.155	0.95	0.00
	Cumulant-ICA (L1)	-0.140	21.690	0.95	0.99	-0.092	64.594	0.95	0.94
	Cumulant-ICA (L2)	0.004	0.120	0.99	1.00	0.004	0.130	0.99	0.94
Aux heteroskedastic	OLS	0.254	0.254	0.00	0.00	0.250	0.250	0.00	0.00
	2SLS oracle	0.005	0.035	0.93	0.00	-0.001	0.016	0.94	0.00
	SP	0.290	0.293	0.00	0.00	0.287	0.288	0.00	0.00
	Lewbel	0.008	0.071	0.94	0.00	-0.001	0.028	0.97	0.00
	Copula CF	-0.030	0.191	0.88	0.00	-0.088	0.106	0.76	0.00
	Cumulant-ICA (L1)	-0.031	4.117	0.99	0.99	-0.012	0.122	0.93	0.39
	Cumulant-ICA (L2)	0.058	0.150	0.93	0.96	0.001	0.101	0.94	0.37

Table 4: Pilot Monte Carlo, S1 versus S3 with $\lambda = 1$. 200 replications per cell; $\beta = 1$; λ as in the text unless fixed by the scenario.

		$n = 1000$				$n = 5000$				$n = 10000$			
Scenario	Estimator	Bias	RMSE	Cov.	NI	Bias	RMSE	Cov.	NI	Bias	RMSE	Cov.	NI
S1 non-Gaussian factor	OLS	0.501	0.502	0.00	0.00	0.499	0.499	0.00	0.00	0.502	0.501	0.00	0.00
	2SLS oracle	-0.002	0.045	0.95	0.00	-0.000	0.020	0.96	0.00	0.002	0.015	0.94	0.00
	Cumulant-ICA (L1)	0.001	0.297	0.92	0.64	0.002	0.100	0.94	0.01	-0.009	0.072	0.92	0.00
S3 Gaussian	OLS	0.499	0.500	0.00	0.00	0.500	0.500	0.00	0.00	0.500	0.500	0.00	0.00
	2SLS oracle	-0.004	0.041	0.98	0.00	0.001	0.019	0.96	0.00	-0.000	0.013	0.95	0.00
	Cumulant-ICA (L1)	0.239	19.704	1.00	1.00	0.182	2.040	1.00	1.00	0.195	6.788	1.00	1.00

Table 5: Pilot Monte Carlo, neural split (60 epochs). 25 replications per cell; $\beta = 1$; λ as in the text unless fixed by the scenario.

		$n = 2000$			
Scenario	Estimator	Bias	RMSE	Cov.	NI
S1 non-Gaussian factor	OLS	0.251	0.251	0.00	0.00
	Cumulant-ICA (L1)	-0.033	0.547	0.96	0.96
	Neural split IV	0.264	0.259	0.00	0.96
	Neural split CF	0.314	0.497	0.88	0.96
S3 Gaussian	OLS	0.251	0.253	0.00	0.00
	Cumulant-ICA (L1)	0.080	3.526	1.00	1.00
	Neural split IV	0.249	0.255	0.00	1.00
	Neural split CF	0.324	0.620	0.76	1.00
Aux heteroskedastic	OLS	0.248	0.252	0.00	0.00
	Cumulant-ICA (L1)	-0.045	0.893	0.92	0.92
	Neural split IV	0.262	0.260	0.00	0.96
	Neural split CF	0.489	0.644	0.48	0.96

(iii) Labelling matters, and (L2) works only where its premise holds. In the Aux design (L2) is median-unbiased with RMSE 0.10 at $n = 5000$. In S1, where $\mathrm{Var}(e_X \mid w_1)$ is constant, the auxiliary slope is uninformative, the rule picks the wrong root about half the time, and the median bias is close to that of OLS. Lewbel's estimator, which relies on the same heteroskedasticity, is excellent in the Aux design and uninformative elsewhere.

(iv) Each competitor wins only in its own setting. SP removes the bias almost completely in S2 (median bias 0.004 at $n = 5000$), where the confounding is a quadratic in x, and coincides with OLS whenever the confounding is linear. In S2 the "oracle" row is not a ceiling: there e_X is itself the confounded variable, which shows that an instrument is only valid relative to a model. The non-Gaussian estimators are misspecified in S2. D flags about 90% of replications, and the median bias of Cumulant-ICA there exceeds 1.6 at $n = 5000$. This calls for the agreement tests of Section 5 as a second line of defence.

(v) Violations of independence (S4) bias every internal-instrument method, including the oracle. Cumulant-ICA is mostly flagged. Across all replications its median bias is small for $\rho = 0.3$ and about 0.28 (against 0.40 for OLS) when e_X has a direct effect on u. This motivates the sensitivity analysis of Conley et al. (2012).

(vi) The unanchored neural split does not identify β (Table 5). The IV version stays within 0.02 of OLS in all three designs, and the control-function version is noisier without being better, as Proposition 4 predicts. Encouragingly, the procedure reports its own failure: 96–100% of replications are flagged as not identified. The anchored split of Section 4.4 is the natural next step.

8 Risks and limitations

- Assumption H1 is equivalent to a disguised exclusion restriction and is not testable. The paper must say so plainly, and results should always be accompanied by sensitivity analysis and by at least one alternative identification strategy.
- Identification through higher-order moments is weak: it requires non-trivial kurtosis (or skewness) and large samples; D makes this visible but does not cure it.
- The label ambiguity of Proposition 1(b) is intrinsic. Assumption H5 is an additional, substantive input.
- Over-complete non-linear ICA is less identifiable than its linear counterpart; without a class restriction the split is arbitrary. Proposition 4 shows that even in the linear case the exogenous source is not a function of the data.
- Novelty risk: the identification layer already exists in the latent-variable LiNGAM literature; the contribution must lie in non-linear estimation, inference and simulation evidence.

9 Next steps

1. Run scenarios S1 and S3 first: if the method cannot separate the identifiable case from the Gaussian one, nothing else matters. The pilot shows that the cumulant layer and the diagnostic D pass this test, while the unanchored network does not improve on OLS.
2. Implement and evaluate the anchored split of Section 4.4.
3. Replace permutation calibration of the residual-independence tests with a bootstrap that re-estimates β.
4. Read Hoyer et al. (2008), Silva and Shimizu (2017) and Tramontano et al. (2025) closely to settle the novelty question.

A Implementation and tests

The accompanying Python package `ivsplit` implements every component of the paper:

Module	Content
`dgp.py`	Source distributions, DGP (3), scenarios S1–S5 and Aux
`estimators.py`	OLS, 2SLS (robust), oracle, Lewbel, copula CF, SP, joint cumulants, Prony roots, labelling, Cumulant-ICA
`independence.py`	HSIC, distance covariance, permutation tests
`split.py`	Neural split, loss (7), sample-split estimation and inference
`diagnostics.py`	The battery of Table 1, Conley et al. bounds, Holm/Bonferroni
`montecarlo.py`	Grid runner and summary metrics

The test suite (`pytest`) checks, among other things: that sources are standardised and have the intended skewness and kurtosis; that the oracle 2SLS is consistent while OLS converges to $\beta + \lambda\sigma_H^2/(\sigma_H^2 + \sigma_X^2)$; that the Prony roots recover $\{\beta, \beta + \lambda\}$ and flipping the sign restriction returns $\beta + \lambda$; that D is large under non-Gaussian confounding and small under Gaussian sources or no

confounding; that SP wins in S2 and fails under linear confounding; that the over-identification test with powers of the instrument detects contamination when $\mathbb{E}[H \mid \tilde{x}]$ is non-linear and is blind when the sources are Gaussian or exchangeable; that HSIC and distance covariance detect non-linear dependence and have correct size; and that the neural split respects the relevance band, uses disjoint halves, and reports failed identification under Gaussian sources.

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